Example 4.40 How many paths are there from start to end on a 6×4 grid as shown in the picture?

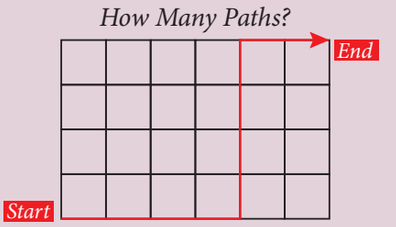


Figure 4.15

Solution:
Note that any such path comprises 6 horizontal unit lengths and 4 vertical unit lengths. This each path consists of 10 unit lengths where 6 are of one kind (horizontal) and 4 are of another kind (vertical).
Thus the total number of paths is $\dfrac{10!}{4! \times 6!} = 210$.

Example 4.41 If the different permutations of all letters of the word BHASKARA are listed as in a dictionary, how many strings are there in this list before the first word starting with B?

Solution:
The required number of strings is the total number of strings starting with A and using the letters A,A,B,H,K,R,S is $\dfrac{7!}{2!} = 2520$.

Example 4.42 If the letters of the word IITJEE are permuted in all possible ways and the strings thus formed are arranged in the lexicographic order, find the rank of the word IITJEE

Solution:
The lexicographic order of the letters of given word is E, E, I, I, J, T. In the lexicographic order, the strings which begin with E come first. If we fill the first place with E, remaining 5 letters (E, I, I, J, T) can be arranged in $\dfrac{5!}{2!}$ ways. On proceeding like this we get,

$E - - - - - = \dfrac{5!}{2!} = 60$ ways
$IE - - - - = 4! = 24$ ways
$IIE - - - = 3! = 6$ ways
$IIJ - - - = \dfrac{3!}{2!} = 3$ ways
$IITE - - = 2! = 2$ ways
$IITJEE = 1$ way

The rank of the word IITJEE is 60 +24 + 6 + 3 + 2 + 1 = 96.

Example 4.43 Find the sum of all 4-digit numbers that can be formed using the digits 1, 2, 4, 6, 8.

Solution:
The number of 4-digit numbers that can be formed using the given 5 digits is ${}^5P_4 = 120$. We first find the sum of the digits in the unit place of all these 120 numbers. By filling the 1 in unit place,